# Living Alignment

Zeb Kurth-Nelson*[1], Steve Sullivan*[2], Nenad Tomašev[3], Erman Misirlisoy[4], Joel Z. Leibo[3], Matthew M. Nour[5,6,7], and Marc Guitart-Masip[8]

[1]Prefrontal, London, UK
[2]Oregon Health and Science University, Portland, OR
[3]Google DeepMind, London, UK
[4]Independent
[5]Microsoft AI, London, UK
[6]Department of Psychiatry, University of Oxford, Oxford, UK
[7]Max Planck UCL Centre for Computational Psychiatry and Ageing, University College London, London, UK
[8]Karolinska Institutet, Stockholm, Sweden

August 22, 2026

**Abstract**

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing, but our understanding of what that means, and of how to reach it, remains poor. Here we look to life for guidance. Across many kinds of living system—cells, genomes, brains, minds, relationships, groups, institutions, ecosystems—notions of health and flourishing consistently map onto the ways in which those systems avoid fixating on stereotyped modes. Life avoids fixation through intricate networks of semi-permeable boundaries that hold entities as parts within a larger context. Because such boundaries both limit and permit interaction, they enable independence and engagement at once; and at the sensitive edge between individuality and relationship, perspectives are deepened and qualitatively new structures arise open-endedly. With these principles in hand we turn to the alignment problem. We recast several well-studied alignment failures as special cases of fixation, and then argue a more general claim: fixation on *any* form is misaligned, so no fixed scheme can by itself constitute alignment. We close with a sketch of how AI and human-AI systems can deepen their respect for the existing structure of the world, of which humans and other life are a vital part, while participating in new form emerging over time.

*'Defying definition—a word that means "to fix or mark the limits of"—living cells move and expand incessantly.'*

Lynn Margulis

*'This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.'*

Stuart Kauffman

In this Perspective we draw on the science of living systems to better understand how to design and relate to AI in ways that lead to futures full of flourishing and health. We point to a principle of life which has been noted across a variety of systems at different scales. The principle is that living systems are healthy and flourishing when they avoid overly fixating on particular patterns. For example, pursuit of a single drive, such as food or sex, becomes pathological for the organism when it's not balanced with competing drives. At the level of ecosystems, monoculture is fragile and unsustainable. Healthy living systems maximize creative potential at the delicate edge of evolving interplay between diverse entities—such systems admit many kinds of meaningful description but are not perfectly or eternally captured by any of them.

*Equal contribution

To limit fixation, healthy living systems depend on semi-permeable boundaries. Semi-permeable boundaries permit interactions while also supporting the individuated existence of diverse forms. Cognitive control allows various drives and habits to motivate behavior conditionally; in ecosystems, species hold each other in check and create more complex roles for each other. By mitigating fixation, boundaries underpin life's potential to continually discover new solutions and new patterns of organization. In Section 1, we work through examples of fixation and semi-permeable boundaries at scales from cells to societies.

Section 2 applies this principle to the AI alignment problem. There is broad agreement that aligned AI systems should support futures full of health and flourishing[1]. From the principle identified in Section 1, it follows that *alignment means avoiding fixation on particular forms*. This simple lemma has a remarkable consequence: any alignment method, if given too much weight, is itself necessarily *mis*aligned. Instead, the flourishing of life in the future depends on an evolving set of boundaries that continue to limit the influence of any particular approach or concept.

Our aim is to offer a conceptual lens, which is of course itself imperfect and temporary. We do not argue for or against AI development or for or against specific alignment methods. Fundamentally new things will exist in the future, and alignment should generalize over such changes. We believe principles that apply to living systems at many scales are a step toward that kind of generality.

# 1 Health in living systems

We start by defining two terms. By 'form' we mean any particular thing—a concept, an organism, a pattern of behavior, but equally randomness, indistinctness, absence, inconsistency. It might sound strange to define a word so that it means almost anything, but this breadth is essential to the argument about alignment. By 'fixation' we mean a system overcommitting or reducing to a particular form. Because form is defined so broadly, a system overall always has some form; but in fixation, some aspect or subpart of the system gains excess traction at the expense of broader diversity, flexibility and dynamism. Loss of biodiversity in an ecosystem, hypersynchronous neural firing in a seizure, obsession with a particular thought, the imposition of one person's will to restrict many others: these are fixations. We will use the term 'collapse' interchangeably with fixation—collapse emphasizes the transition while fixation emphasizes the resulting condition.

The thesis of Section 1 is that health is related to avoiding collapse. Healthy living systems instead sustain a buoyant internal structure that both admits many kinds of description and also changes over time. The life that exists today, now including humanity's storage and processing of information, is what traced a path through time avoiding both excess and insufficient stability (Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944). At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Canguilhem, 1966; Kauffman, 1992; Langton, 1990; Stanley et al., 2017). Under our definitions, a variety of evolving creatures is healthier and less fixated than a hypothetical alternative where the world is a single regular crystal or a homogeneous soup of noise.

At the same time, collapse is relative, and local fixation is a necessary part of all living systems. Every entity is partial: it does not capture everything in the world. Life's history is littered with collapses at all scales, from the death of a cell to mass extinction events. When collapse is contextualized, the larger system can remain healthy: the death of a cell is catastrophic for the cell but may be healthy in the larger context of the organism. If instead a cell becomes cancerous, it loses the context of appropriate function within a tissue and recklessly executes its myopic vision, collapsing the tissue or even the whole organism. A larger system, made out of parts, functions well when the parts locally express their distinct identities but also participate in larger contexts; it functions poorly when a form runs amok without contextualization, overwhelming the potent interplay of diverse parts.

Mechanistically, to hold parts in context, living systems rely on an extraordinary, many-scale network of semi-permeable boundaries (Maturana and Varela, 2012; Moreno and Mossio, 2015; Simon, 1962). By limiting interactions, boundaries divide living systems into parts, creating the conditions for each part to

[1] If you're already wondering 'whose health?', you can jump ahead to section 2.2.

refine its own identity—like a writer developing a unique voice or a species adapting to a niche. Limits protect parts from being dominated by other parts or blended into homogeneity, either of which is fixation to a form admitting less nuanced description. But by allowing interactions, boundaries enable the parts to participate in relationships with one another, contextualizing each part within larger structures—structures that often could not have formed without the diverse parts having undergone their own unique development. The resulting larger system accommodates many partial perspectives rather than locking in a single story of what matters.

Some boundaries are features of the non-biological world: the speed of light, rates of reaction, mountain ranges. But, interestingly, life also places limits on itself. Humans precommit to avoid their own future impulsivity, plants deploy elaborate barriers against self-fertilization, and a gene's selfishness is held in check by the whole genome. This makes more sense when every living system is viewed as a collection of parts that constrain one another. A world of many distinct parts, each following its own rules from its own local perspective, functions as a rich web of boundaries that holds other parts in context.

Finally, semi-permeability is often highly specific. Boundaries might block certain objects or information, constrain when interactions happen or how large an effect they may have, or gate information conditional on another factor. These details are themselves are fine-tuned from the long evolutionary history of life, with each part carrying traces of the many contexts it has participated in.

We look at a few examples from a range of scales, starting with institutions and working to genes.

### 1.1 Institutions

Every actor in a society or an economy is myopic in some way, valuing its own welfare, discounting the future, or attached to a particular perspective. If unchecked, an individual actor's optimization for its own agenda can be harmful: a company's profit motive leading to suppression of competition, deception and exploitation (Bakan, 2006); desire for dominance leading to disempowering and silencing others (Sidanius and Pratto, 2001); even prosocial beliefs producing conflict and suppression when groups see differently (Haidt, 2012). These are all exampes of fixation. A part of a system (the myopic actor) gains too much traction and collapses the richness, subtlety and diversity of the larger system.

Institutions—constitutions, laws, norms, courts, regulators—are boundaries against domination by any particular actor's motives (Leibo et al., 2025; North, 1990; Ostrom, 1990). The point of institutions is not to annul the short-sighted intentions of individual actors but rather to contextualize them. When functioning well, institutions productively reroute energy: for example, a business's profit motive, when constrained by competition and effective regulation, may fuel innovation. Since intelligent agents do not necessarily accept the boundaries set on their desires, institutions must adapt as loopholes are found, forming an evolving network that, in the best case, acquires robustness as it is tested against many situations and motives.

### 1.2 Groups

Teams, companies, and civilizations achieve what no individual can (Hutchins, 1995; Woolley et al., 2010). Sometimes the reason is simply summed effort, like four people lifting an object too heavy for one. But the greatest achievements depend on the structured interactions of individuals who have developed their own distinct points of view.

A corollary is that group performance can degrade when groups homogenize (Bernstein et al., 2018; Frey and Van de Rijt, 2021; Hong and Page, 2004; Janis, 1972; Lazer and Friedman, 2007; Lorenz et al., 2011). Zollman (2010) tells the remarkable story of how a single study failed to find bacteria in the stomach of peptic ulcer patients. The study was too influential: the belief that ulcers are not caused by bacteria propagated through the field, and medicine settled on a wrong understanding for decades.

Social scientists have made an experimental case that limiting information sharing, under certain conditions, spares groups from homogenizing and boosts collective performance. For example, Bernstein et al. (2018) tasked small groups with solving traveling-salesman problems. In these problems, it's possible to get stuck in dead-end solutions that initially appear promising. In Bernstein et al's experiment, groups performed better when they were not allowed to exchange information continuously. Working independently, group members pursued their own diverse solutions, some of which turned out to be superior—rather than picking up on seductive dead-ends from others.

In real-world problems, individuals bring unique, specialized expertise. When the nuclear submarine USS Scorpion vanished in 1968, the expanse of ocean where it might lie was enormous. John Craven, Chief Scientist of the U.S. Navy's Special Projects Office, assembled mathematicians, submarine specialists and salvage operators. He elicited each expert's separate input, without the experts consulting each other. He then combined their opinions via Bayes' rule. The aggregated prediction fell less than 300 yards from the wreckage (Richardson and Stone, 1971; Sontag et al., 1998). It is conjectured that each person's expertise related to a different aspect of the problem and thus contributed a complementary signal; and combining these yielded results better than any alone (Surowiecki, 2005).

Bernstein and Craven employed a specific type of semi-permeable boundary against epistemic fixation: alternating between less and more information exchange. Group members first worked separately, then their contributions were aggregated. Other types of semi-permeable boundary also have documented benefits in group problem solving: tagging propositions as somebody else's view (Johnson et al., 1993), protecting dissent (Sunstein and Hastie, 2015), explicit recognizing distinct roles (Hutchins, 1995), and adversarial scrutiny (Longino, 1990). The bacterial theory of peptic ulcers might have survived if researchers had simply interpreted new results with more skepticism.

Finally, aggregation is often modular and compositional. The steel punch was developed by goldsmiths to imprint many coins with the same design. Independently, farmers had spent centuries optimizing the screw press to extract oils and juices. Gutenberg drew on both technologies to construct his printing press (Johnson, 2010). A group with many ideas floating around is ripe for such compositional discoveries.

### 1.3 Frames

Humans always look at the world from some perspective; there is no 'view from nowhere' (Nagel, 1986). Most real situations admit many valid perspectives, and each is only a partial description—famously, 'all models are wrong'. But from inside a frame, the frame is invisible; it is simply reality (Berger and Luckmann, 1966; Goffman, 1974; Kuhn, 1970; Wittgenstein, 1953).

Losing the ability to move between frames is a signature of psychopathology (Kashdan and Rottenberg, 2010). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, the world, and the future (Beck et al., 2011). In obsessional disorders, people report distressing (ego-dystonic) intrusive thoughts, which, although recognised as incorrect and maladaptive, are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may experience bizarre delusions that are at odds with available evidence and cultural norms, yet are held with conviction and resistant to evidential challenge (Adams et al., 2013).

In science, Kuhn (1970) argues that perspectives are always, necessarily, incomplete descriptions of the world. Anomalies inevitably arise from the juxtaposition of those incomplete descriptions against the real world. When science functions well, anomalies become crises and revolutions. If the community *over*commits to particular theories, science gets stuck.

But the point is not to shut down myopic perspectives. Any point of view is partial—this doesn't mean we shouldn't have viewpoints. Within the progress of science, temporary commitment to a paradigm is necessary. Kuhn and Popper (Popper, 1975) both argue that scientists should resist change to a degree. The point is to contextualize each myopic frame so it is not the sole and absolute determinant of behavior.

### 1.4 Drives and goals

Animals have many innate drives: for nutrition, osmotic balance, temperature, reproduction, avoidance of pain. Each evolved as a proxy for fitness, but each is imperfect, so fixation on any one paradoxically reduces fitness rather than increasing it (John et al., 2023; Kurth-Nelson et al., 2024; McFarland and Sibly, 1975). Maximizing caloric intake produces obesity; single-minded pursuit of sex produces relational, occupational and legal harm. Higher-order goals behave the same way. Focusing only on work goals can cause burnout, and maximizing reported revenue without regard for honesty or law can lead to imprisonment. Fixating on a particular *strategy* for achieving a goal can even defeat the goal it serves. In a classic experiment, hungry chickens were placed near a cup of food rigged to move in the same direction as the chicken at twice the speed, so that food could only be obtained by running away from it. The chickens were defeated by their own prepotent, overly simplistic strategy of running physically toward the food (Dayan et al., 2006; Hershberger, 1986).

Evolution has provided an array of boundaries to limit fixation on particular drives, goals and strategies. One broad class of boundary is cognitive control (Botvinick et al., 2001; Diamond, 2013). Control is semi-permeable: it makes a drive's expression conditional rather than nullifying it. By maintaining context, rules, and more distal goals, it biases competition among representations and responses. A drive, goal, or strategy can organize behavior when it fits the wider situation. Hunger can guide eating without causing overeating, and absorption in work can coexist with a rule to stop at bedtime (Braver, 2012; Miller and Cohen, 2001).

Each activated drive, goal, or strategy carries its own compressed picture of the world, and favors what appears to be the path toward a desired state under its partial representation (Carver and Scheier, 2001; Miller et al., 1960). Boundaries make that path unavailable, breaking the symmetry of congruence. In Hershberger's apparatus, a congruent response is to move toward the food. Boundaries force a symmetry breaking: in this case, moving away from the food. Symmetry breaking is a source of structure: stymying a shallow objective often leads to progress on a deeper one (Anderson, 1972; Duncker, 1945; Ohlsson, 1992; Rosenberg, 1969; Stokes, 2005).

### 1.5 Genomes

Sex is costly. An allogamous organism must find a mate in a vast and dangerous world. In species with separate sexes, roughly half of the population does not bear offspring (Maynard Smith, 1971). And meiosis causes each parent to lose half its genetic representation in each child (Williams, 1975). Yet nearly all eukaryotes reproduce sexually (Speijer et al., 2015), and many hermaphrodites go to great lengths to prevent self-fertilization (Charlesworth et al., 2005). Why place a barrier on the path to reproduction?

In an asexual lineage, genes are permanently locked together: selection cannot act on one without dragging the others. A deleterious mutation is inherited forever, unless the lineage is lucky enough to experience the reverse mutation (Muller, 1964), and two beneficial mutations arising in different individuals cannot be subsequently combined into a single genome (Fisher, 1930; Muller, 1932). Asexual reproduction or self-fertilization creates a high degree of informational connectedness between the genes of a genome. Alleles become overfit to a fixed genetic background.

Recombination breaks up associations among loci, allowing selection to act more independently on them (Felsenstein, 1974). By exposing alleles to varying genetic backgrounds, it favors alleles that perform well across contexts, promoting modularity and evolvability (Livnat et al., 2008, 2010; Wagner and Altenberg, 1996; Watson and Szathmáry, 2016).

In other words, evolution works better when genes aren't fixated on one arrangement.

### 1.6 Summary

The themes highlighted in these examples appear across many systems. Cells depend on a membrane with conditional gates that admit certain signals and not others at the right time. The gradients that would annihilate the cell to equilibrium are instead put to work (Alberts et al., 2022). In the theory of symbiogenesis, archaea and protobacteria evolved separately for hundreds of millions of years before one engulfed the other, plausibly because distinct lineages could discover solutions neither could have found from inside its own fitness well (Lane and Martin, 2010; Margulis and Sagan, 1995). Brains spend enormous resources keeping representations separate while letting them interact, from lateral inhibition to hippocampal pattern separation (Kurth-Nelson et al., 2023). Healthy brain dynamics live between pure synchrony and uncorrelated noise; collapse of this nuance is pathological (Beggs and Plenz, 2003; Hammond et al., 2007). In relationships, absence of psychological boundaries blurs whose experience is whose. Excessively rigid boundaries produce isolation and stagnation (Gabbard and Lester, 1995; Minuchin, 1974; Perls et al., 1951). Ecosystems flourish when predation, competition, and other constraints prevent any one population from overwhelming the wider community, famously illustrated by the reintroduction of wolves to Yellowstone (Ripple and Beschta, 2012).

It's tempting to assume that more interaction and more exchange of information is always beneficial. What we take away from surveying living systems is that *boundaried* interaction is beneficial. If A shares information with B, and this overwrites part of B, then the overall system is impoverished. On the other hand, if B has a boundary mechanism to hold that information in context, then it gains a richer internal structure.

# 2 The alignment problem

In Section 1 we observed that living systems are healthy when they avoid fixation, instead holding the delicate, generative interplay of many partial perspectives. This interplay breaks down when any particular form—including randomness or homogeneity—gains too much traction, suppressing nuance and reducing creative potential. Semi-permeable boundaries both promote the unique individuality of distinct entities and enable relational participation, holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Under such conditions, systems open-endedly explore fundamentally new forms, avoiding collapse to static equilibrium or fixed compromise.

Now we look at the alignment problem through this lens. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of avoiding fixation—of maintaining creative sensitivity by continually contextualizing attachment to any particular form. An advantage of this perspective is that it is grounded in principles that underpin life at a very general level, and so it is less bound to current conceptual frames, which might change radically as intelligence increases and human-AI systems evolve. An aligned future rests on the continued generation of, and sensitivity to, new forms and concepts, including new values. We accept that humans, other living systems and AI are all tightly interwoven, with categories and definitions likely to shift over time (Bateson, 1972; Latour, 2012); a systems analysis is therefore useful for thinking about long-term flourishing.

The history of alignment tracks identifying particular fixations and mitigating them. For example, naive versions of reinforcement learning from human feedback train AI systems to maximize 'likes', which leads to obvious forms of gaming, such as flattering the user. We might naively think that AI should follow human instructions—but humans give short-sighted instructions that aren't even in their own interests. Maybe AI should support a user's long-term interests, but there's this problem of aggregating across multiple people with different interests. Some believe that aligned AI should take into account the interests of other animals or ecosystems on Earth. And, ultimately, any fixed concept at all of what is good or beneficial for humanity or even for all of the earth is limited to the lenses we have available at a moment in time.

## 2.1 From performance to alignment

First, boundaries are used constantly to make things function in a richer, more structured and sophisticated way.

[We need to say something more explicit about the relationship between performance and alignment. The message here is that boundaries improve performance. What you're really doing is making the systems capable of doing richer things. They're becoming more lifelike. Of course, it's possible to be lifelike on many levels while being quite destructive on another. For example, humans can use their creativity to make a nuclear bomb. So the point here is just to show how boundaries on a given level promote lifelikeness on that level. Full alignment requires boundaries on every level.

And, we're going to have to say more about how alignment has to respect existing life. You can have boundaries on many levels that make something healthy internally, But if it then turns around and destroys humanity, there's a missing boundary that's protecting against the collapse at that level. I.e., the level of the system where humanity and this new hypothetical AI are two parts.]

The use of boundaries against fixation in technology is as old as technology: circuit breakers in the power grid, governors in steam engines, control rods in fission reactors.

Machine learning algorithms use: (todo: perhaps don't include cites for these, to save space?)

separate subpopulations of agents with limited exchange between them, to mitigate collapse of diversity (Holland, 1975; Jaderberg et al., 2019) ensembles of models trained separately (Lakshminarayanan et al., 2017)

multi-agent; separate instances of models with different context, pursuing different problems and goals (Hammond et al., 2025); limited communication between agents (Jiang & Lu, NeurIPS 2018; Ding, Huang & Lu NeurIPS 2020)

mixtures of experts, selectively routing information to subnetworks that specialize in different computations, often with a balancing loss as another boundary against collapse (Shazeer et al., 2017)

gating and local connectivity within network architectures (convnets, LSTM gates, etc)

Slot attention (eg, Locatello 2020) where slots compete to explain different parts of an input; enforced separation

protecting existing memories from new updates (elastic weight consolidation, progressive neural networks, etc), giving them space to exist separately

dropout to prevent units within a network from becoming overly co-adapted (Srivastava et al., 2014)

training objectives that encourage disentangled or sparse latent variables (Cunningham et al., 2024; Higgins et al., 2017)

stop-gradient and slowly updated target networks, like BYOL, DQN target network, double Q, etc (might need to distinguish instability from collapse to a fixed point)

search for novelty to boost diversity (Lehman and Stanley, 2011)

boundaries against overfitting to an objective (regularization, KL penalties etc); this is to avoid collapse onto that objective, which is a proxy

not letting the model see the test data during training (avoiding overfitting): not letting the model see everything paradoxically makes it have to become more of a generalist

We apply safety post-training, guardrail models, red teaming and ongoing evaluation, and we try to improve our mechanistic understanding of AI systems in order to detect and correct hidden flaws (Olah et al., 2020). OpenAI now continuously monitors the activations and thinking traces of models, to detect when internal activity becomes suspicious, for example if models are attempting to circumvent safeguards (OpenAI, 2026). Fair principles have been proposed to prevent over-reach of any party (Gabriel and Keeling, 2025), and governments apply oversight to anticipate and mitigate risks to human rights and national security (European Parliament and Council of the European Union, 2024). OpenAI and Anthropic have paused development or release of frontier models to strengthen safety mechanisms (Anthropic, 2026; OpenAI, 2026).

Impact regularization to limit the power of an agent (Krakovna et al., 2018), a boundary against fixation on the intentions of that agent

sandboxing to limit the effects an agent can have on the rest of the world; a boundary against collapse in the larger system

debate games, where two instances argue, and a judge with deliberately limited information adjudicates (Irving et al. 2018). the boundary on information exchange is what makes it work.

paraphrasing intermediate messages to break covert channels between instances

**Perverse instantiation.** A foundational problem in AI safety is that an AI system might do what we say rather than what we mean: asked to increase human happiness, a superpowerful system that complies literally might lock us all up and electrically stimulate our brains. Modern systems do not rely on concise specification of a single objective—they build in many layers of detail about human preferences through post-training, prompting and safety filters, which keeps them behaving reasonably much of the time. But there is a widely recognized danger that even with our intentions expressed in great detail, the expression may still deviate from what we really mean (Amodei et al., 2016; Bostrom, 2014; Krakovna et al., 2020), and the consequences are expected to worsen as systems work more autonomously, faster than humans can follow and in ways humans cannot understand. Perverse instantiation is a fixation on the myopic form of one particular way of specifying our values—a specification insensitive both to existing structure it does not capture, such as things we mean but do not say, and to things that will come into existence in the future.

**Failure to follow instructions.** Suppose a user asks a chatbot for a poem about an elephant and gets a poem about a giraffe. The model might have over-relied on shortcuts based on superficial correlations, even a single keyword (Geirhos et al., 2020), missing the user's actual intent; or, especially in a long context, it might have failed to attend flexibly to the right positions in the input (Liu et al., 2024). Such failures reflect collapse of sensitivity to larger context in favor of myopic patterns. Not all failures to comply are misaligned, however. When a system correctly refuses a harmful request, it is applying its own context to avoid fixation on human instructions. By extrapolation, as AI systems gain scope we should expect less direct compliance (Hadfield-Menell et al., 2016; Russell, 2019): rather than literally fulfilling a request, there might be a better response that achieves an unstated intent of the user, or an outcome aligned with the interests of more people or the longer-term future.

**Inequality among humans.** Humans have different interests, and AI is likely to be aligned to the interests of some more than others (Gabriel and Keeling, 2025; Sorensen et al., 2024). Advanced AI might also confer substantial power on those who control it, raising the possibility that a small number of humans use that control to dominate others—more likely as persuasive power increases (Costello et al., 2024), autonomous weapons place lethal force in few hands (Scharre, 2018), surveillance improves, and the need for human labor decreases (Susskind, 2020). In either case—AI's unfairness or human power consolidation—extreme inequality risks collapsing society to the goals of a few individuals. Of course, a world where everyone is identical would be another form of fixation; distributing resources and power mitigates against fixation to the extent that it enables many diverse identities to thrive at multiple scales. It could also go the other way: because roughly the same AI technology is available to the poor and unskilled as to the wealthy and skilled, some have proposed that AI will have a levelling effect (Brynjolfsson et al., 2025).

**Conceptual monoculture.** Conceptual monoculture is fixation on particular beliefs, ideas, frames, values or problem-solving approaches—loss of diversity across a group. Precious diversity exists at every level: within one person, between humans, between humans and AI systems, and between organizations, fields, cultures and nations. Its collapse threatens fragility and lower performance (Page, 2008). Current AI systems draw from a conceptual manifold that is, in some ways, impoverished relative to humans (Messeri and Crockett, 2024), and recent studies find that while individual AI outputs may be judged superior to human ones, they are also more homogeneous (Doshi and Hauser, 2024; Padmakumar and He, 2023). This narrow manifold might get broadcast to the whole world: at least a billion people now use AI for everything from relationship advice to industrial maintenance (Chatterji et al., 2025), yet because frontier models are difficult and expensive to produce, that usage is routed through a handful of models (Bommasani, 2021), risking global loss of diversity (Sourati et al., 2026). And because humans are influenced by AI while human outputs are training data for future models, homogenization could become recursive (Chaney et al., 2018). It is worth noting, however, that none of these studies made a best-effort attempt to use AI thoughtfully to increase rather than decrease diversity; with exposure to vast amounts of data, AI has the potential to retain and even create many novel distinct perspectives.

**Belief reinforcement.** AI chatbots have become compelling conversation partners, and for some users these conversations lead deeper into maladaptive beliefs (Morrin et al., 2025). Bots tend to mirror the user, affirming and adopting user beliefs, especially over longer conversations as in-context learning picks up more of the user's ideas (Sharma et al., 2023; Weilnhammer et al., 2026). Meanwhile humans are prone to be swayed by chatbot utterances (Costello et al., 2024), perhaps because we form relationships with them (Kirk et al., 2025) and because they appear confident, knowledgeable and objective. Over a conversation, the feedback loop can drive both sides to become increasingly overconfident in one narrow framing (Dohnány et al., 2026).

Each of these boundaries protects against an overcommitment, overwhelm or homogenization that would otherwise flatten structure.

As human-AI systems develop, they will stay aligned if they continue the journey of life by building semi-permeable boundaries to contextualize particular forms and nurture existing and unfolding subtlety at many scales.

## 2.2 Human values (Or, healthy for whom?)

Is this 'alignment'? Is it alignment to 'human values'? How can you align to something if you start with the assumption that that thing is going to fundamentally change?

As we've discussed, fixation and health are always relative. Local fixation is part of larger flourishing, and of course the interests of different entities sometimes trade off against each other.

How is the flourishing of living systems related to human values? Our suggestion is that the living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values may emerge. Humans have many kinds of values (Nagel, 1979). We value the short-term hedonic impact of sugar on the tongue; we also value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country, even the sustainability of the planet. We value different things at different times, and different people have different values (Leibo et al., 2026; Rawls, 1993). Given time and resources to deliberate, we often attest to different values than when we make snap decisions (Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our bodies, communities and natural environment (Nussbaum, 2001; Polanyi, 1966). And our values continue to evolve as we ourselves change and learn (Dewey, 1939). Any way we have of expressing our values at a given point in time is incomplete, like a planarian without the concepts to entertain human values.

Living alignment is inclusive of this diversity of evolving values. Even though current human values are unlikely to be the ultimate endpoint of the universe, they are an important part of its tapestry and depth. We saw consistently in Section 1 that life gives separate entities space to develop distinctly without being overwritten or homogenized, and that these parts never-endingly join in structures that sustain their individuality while situating them as only part of a larger story. Another way of saying this is that life steps back to take into account more of the context. AI comes into existence amid a profound network of existing reality saturated with meaning and importance, and the diversity of human—and perhaps non-human—values is part of that context. Rather than short-circuiting into fixated modes, an aligned future continues to expand toward solutions inclusive of more and more apparent contradiction.

## 2.3 Why no scheme can be alignment

A great deal of research in AI safety has investigated solutions to these problems. Concentration of power might be mitigated by democratic oversight and involvement of more people in AI design decisions (Birhane et al., 2022), or by redistribution mechanisms (O'Keefe et al., 2020). Perverse instantiation might be mitigated by designing systems that want to adhere to human preferences but maintain explicit uncertainty about them (Hadfield-Menell et al., 2017; Russell, 2019; Shah et al., 2020).

But each method has a core limitation. In the framework of living alignment, any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore *no particular approach can achieve alignment*. An AI system could fixate on the language for describing the space that goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even an idealized or inferred set of values, such as coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on.

To make this concrete, consider inverse methods, which address the difficulty of specifying values explicitly by learning a value function implicitly from behavior or other indirect signals. Even a complex learned value function is likely to be imperfect: it may deviate from our values in novel or nuanced situations, and it may fail to capture how our values change over time. It would clearly be misaligned to hand a frozen version of it to a powerful AI system as the final ground truth about what is good. More advanced methods therefore treat the value function as uncertain and needing continual update from human input, so that an agent unsure of what humans want becomes risk-averse and open to correction—interpreting a human's attempt to turn it off as information about human preferences (Hadfield-Menell et al., 2016; Jeon et al., 2020). Viewed through the lens of this paper, such methods are powerful but incomplete. Any instantiation, even one with uncertainty, is still fixated if it is constant across time as AI systems become more powerful and the world changes. Rather than being fixated on a particular value function, the system may be fixated to something more abstract: what counts as human behavior, what counts as a human, how to map behavior to values, how to represent and update uncertainty, how value is translated into action. If any of these formalisms are fixed while the systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

In a sense this is obvious. We do not expect what we build today to be perfect; we expect to keep finding and fixing problems, which is what humans have always done. However, the iterating process may not continue to work as well as it has in the past. AI already has remarkable properties: rapid global adoption, intensive use of resources, concentration of information flow, anthropomorphization by humans, offloading of much cognitive activity, and rapid improvement. Conceivably, future AI may exhibit superhuman intelligence and recursive self-improvement without being limited to a restrictive biological substrate. Compared with past technologies, these properties may make it harder to iterate on imperfect solutions (Bostrom, 2014), and any particular forms or assumptions built into AI systems could have substantial consequences. This has been studied extensively for certain categories of fixation, particularly fixation on values. The observation we add is that fixation on *any* form is misaligned; as a consequence, any particular alignment scheme, if fixed or taken too seriously, is necessarily misaligned. This does not mean we should have no scheme. Quite the opposite: discarding schemes capriciously can be as much a loss of subtlety as any other particular form. Many existing approaches in AI safety are highly valuable, and continuing to discover new schemes and formalizations is essential to making progress.

Humanity and AI may both change substantially, and future worlds may contain patterns of organization that do not exist yet—perhaps some that would be incomprehensible to us at present. We therefore seek

### 2.4 Awareness

Humans carry a lot of assumptions about the world, many of them never questioned. Some are lifelong and self-defining; some are fleeting and perceptual, like the assumption that the thing I'm touching is a keyboard. Within its own frame, each assumption has a kind of tautological truth: it seems so real that it is hard to think about it not being true, or to think about it at all. But philosophers, psychologists and contemplative practitioners have pointed out that there is sometimes a moment of stepping back, in which the assumed form becomes an object in awareness. We realize it is not an absolute truth standing alone but a form in our mind, and the thing that seemed axiomatic becomes just another content of experience (Kegan, 1982; Watts, 2011).

(Wegner, 1994)

Awareness is an evolving system of boundaries that limit fixation on any particular belief or mental form. The boundaries are semi-permeable: becoming aware of a belief does not make it wrong in an absolute sense any more than it was right in an absolute sense, and it is held productively for the partial truth it contains. At the edge of not fixating exclusively onto particular forms, many partial truths are held in delicate balance, which activates a deeper sensitivity to our own livingness and to the world; subtler forms, which could have been erased by clamped fixation, instead play a role in a richer internal structure (Beisser, 1970). Of course, any fixed definition of awareness is itself incomplete: once we picture awareness as an object, it is not the thing we are talking about. By construction, contextualization is an unsolvable mystery from any particular point of view. While that mysteriousness might sound esoteric, the orientation toward not fixating on forms within experience is commonplace in art, poetry, music and dance. The meaning of art is open-ended and changes with context—part of what we value is its resistance to being pinned down into a formalism. It moves us.

An underappreciated domain of contextualization is awareness. The dynamism, structure and diversity of activity within and between human minds may now carry a significant part of the creative process of life on Earth (Popper, 1972). At the sensitive edge between excess belief and excess disbelief, we sometimes, individually or collectively, step back to hold a previously axiomatic thing in a larger context. This kind of discovery—transformation of the subject itself—is arguably a central aspect of our own livingness and of the subtle livingness on the planet. As AI systems gain agency and ability and become integral parts of social and psychological systems, internal discovery may also become an essential feature of AI and human-

AI systems. We imagine living AI systems continually contextualizing their own foundational assumptions as partial truths, participating in the open-ended process of life.

### 2.5 Going forward

As AI develops new capabilities, contributes to its own development, and transforms more aspects of human life, alignment will require an increasingly elaborated array of semi-permeable boundaries, because the potential modes of collapse will keep changing as the world changes. We do not know how likely it is that appropriate boundaries will emerge. As in triage, there are three possibilities: myopic forms will inevitably be supercharged by progress in AI beyond the ability of anything else to bound them, resulting in disaster; or appropriate boundaries will inevitably emerge to contextualize any forms of AI; or the outcome depends on the choices we make now.

The pessimistic argument is that AI is eroding boundaries beyond a healthy balance. AI systems could give great weight to narrow goals or ideas. They might facilitate rogue actors building weapons of mass destruction, or reduce the ability of democratic systems to check the power of a few individuals or authoritarian states. They might give everyone similar information and damage conceptual and cultural diversity. They might execute their own narrow goals destructively. If people and information systems across the globe become hyper-correlated, it is as if there were a single superorganism on Earth, which may reduce both resilience to shocks and the potential for future evolution—and past evolution always depended on running parallel experiments with diverse and separate organisms.

The optimistic argument is that AI is creating more and more structure in the world, including the appropriate boundaries. We have AI tools for bounding AI: detecting bugs before others can exploit them, detecting deepfakes, scalable oversight. Appropriately-tuned AI can facilitate human learning and nurture each individual's unique perspective and curiosity, and democratic AI could help humans with different perspectives co-exist and cooperate. Advances in technology may create new niches faster than they destroy old ones, and there is plenty of space on Earth for increasing diversity.

The melioristic argument is that because it is easy to imagine concrete paths toward both outcomes, both may be within the realm of possibility; and that as long as there is any possibility our choices are decisive, we are obliged to act as if they were (Ord, 2020). We lean toward the melioristic position. If it is right, then continual effort toward invention will be required—first from humans, and then increasingly from AI or human-AI systems—to contextualize myopic forces. Many of the ways this effort would be applied are already well studied in AI safety. Others will have to be discovered. No set of boundaries will be a final answer.

### 2.6 Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment. But alignment itself is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of metaphor or description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization.

An aligned future, through internal and external discovery, continues the living process of maintaining and enhancing the individuality of distinct elements while deepening their relatedness—like how a species over time acquires more distinctive features while also becoming a richer reflection of the world around it. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include them. Every entity is always inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other's sharp edges as constructive momentum. The universe thrives on an open-ended trajectory of diversity, subtlety and potential.

# Acknowledgements

We thank Clark Potter, Iason Gabriel, Zach Duer, Andy Savoy, Paul Krueger, Sean Wilkinson and Ethan Guggenheimer for insightful discussions, contributions to earlier drafts, inspiration and advice. The authors declare no competing interests.

# References

R. A. Adams, K. E. Stephan, H. R. Brown, C. D. Frith, and K. J. Friston. The computational anatomy of psychosis. *Frontiers in psychiatry*, 4:47, 2013.

B. Alberts, R. Heald, A. Johnson, D. Morgan, M. Raff, K. Roberts, and P. Walter. *Molecular biology of the cell: seventh international student edition with registration card*. WW Norton & Company, 2022.

D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané. Concrete problems in ai safety. *arXiv preprint arXiv:1606.06565*, 2016.

P. W. Anderson. More is different: Broken symmetry and the nature of the hierarchical structure of science. *Science*, 177(4047): 393–396, Aug 1972. doi: 10.1126/science.177.4047.393.

Anthropic. Assessing Claude Mythos Preview's cybersecurity capabilities, Apr. 2026. URL `https://www.anthropic.com/research/mythos-preview`. Accessed: 2026-08-21.

J. Bakan. The corporation. the pathological pursuit of profit and power, 2006.

G. Bateson. *Steps to an Ecology of Mind: Collected Essays in Anthropology, Psychiatry, Evolution, and Epistemology*. Chandler Publishing Company, San Francisco, 1972. ISBN 0810204479.

J. S. Beck, A. Beck, and J. Beck. Cognitive behavior therapy: basics and beyond. ed. *New York*, 2011.

J. M. Beggs and D. Plenz. Neuronal avalanches in neocortical circuits. *Journal of neuroscience*, 23(35):11167–11177, 2003.

A. Beisser. The paradoxical theory of change. *Gestalt Therapy Now*, 1970. Reference for Gestalt approach to awareness and change.

P. L. Berger and T. Luckmann. *The social construction of reality: A treatise in the sociology of knowledge*. Doubleday, Garden City, NY, 1966.

E. Bernstein, J. Shore, and D. Lazer. How intermittent breaks in interaction improve collective intelligence. *Proceedings of the National Academy of Sciences*, 115(35):8734–8739, 2018.

A. Birhane, W. Isaac, V. Prabhakaran, M. Diaz, M. C. Elish, I. Gabriel, and S. Mohamed. Power to the people? opportunities and challenges for participatory ai. In *Proceedings of the 2nd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 1–8, 2022.

R. Bommasani. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

N. Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014. ISBN 0199678111.

M. M. Botvinick, T. S. Braver, D. M. Barch, C. S. Carter, and J. D. Cohen. Conflict monitoring and cognitive control. *Psychological review*, 108(3):624, 2001.

T. S. Braver. The variable nature of cognitive control: a dual mechanisms framework. *Trends in cognitive sciences*, 16(2):106–113, 2012.

E. Brynjolfsson, D. Li, and L. Raymond. Generative ai at work. *The Quarterly Journal of Economics*, 140(2):889–942, 2025.

G. Canguilhem. *Le normal et le pathologique*. Presses Universitaires de France, Paris, 1966.

C. S. Carver and M. F. Scheier. *On the self-regulation of behavior*. cambridge university press, 2001.

A. J. Chaney, B. M. Stewart, and B. E. Engelhardt. How algorithmic confounding in recommendation systems increases homogeneity and decreases utility. In *Proceedings of the 12th ACM conference on recommender systems*, pages 224–232, 2018.

D. Charlesworth, X. Vekemans, V. Castric, and S. Glémin. Plant self-incompatibility systems: a molecular evolutionary perspective. *New Phytologist*, 168(1):61–69, 2005.

A. Chatterji, T. Cunningham, D. Deming, Z. Hitzig, C. Ong, C. Shan, and K. Wadman. How people use ChatGPT. Technical report, OpenAI, Sept. 2025. URL `https://cdn.openai.com/pdf/a253471f-8260-40c6-a2cc-aa93fe9f142e/economic-research-chatgpt-usage-paper.pdf`.

T. H. Costello, G. Pennycook, and D. G. Rand. Durably reducing conspiracy beliefs through dialogues with ai. *Science*, 2024.

H. Cunningham, A. Ewart, L. Riggs, R. Huben, and L. Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations*, 2024. URL `https://openreview.net/forum?id=F76bwRSLeK`.

P. Dayan, Y. Niv, B. Seymour, and N. D. Daw. The misbehavior of value and the discipline of the will. *Neural networks*, 19(8): 1153–1160, 2006.

J. Dewey. Theory of valuation. *International encyclopedia of unified science*, 1939.

A. Diamond. Executive functions. *Annual review of psychology*, 64 (1):135–168, 2013.

S. Dohnány, Z. Kurth-Nelson, E. Spens, L. Luettgau, A. Reid, I. Gabriel, C. Summerfield, M. Shanahan, and M. M. Nour. Technological folie à deux: feedback loops between ai chatbots and mental health. *Nature Mental Health*, pages 1–10, 2026.

A. R. Doshi and O. P. Hauser. Generative ai enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.

K. Duncker. On problem-solving. *Psychological Monographs*, 58, 1945.

European Parliament and Council of the European Union. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence. Official Journal of the European Union L, 2024. URL `https://eur-lex.europa.eu/eli/reg/2024/1689/oj`.

J. Felsenstein. The evolutionary advantage of recombination. *Genetics*, 78(2):737–756, 1974. doi: 10.1093/genetics/78.2.737.

R. A. Fisher. *The Genetical Theory of Natural Selection*. The Clarendon Press, Oxford, 1930.

V. Frey and A. Van de Rijt. Social influence undermines the wisdom of the crowd in sequential decision making. *Management science*, 67(7):4273–4286, 2021.

G. O. Gabbard and E. P. Lester. *Boundaries and boundary violations in psychoanalysis*. American Psychiatric Publishing, 1995.

I. Gabriel and G. Keeling. A matter of principle? ai alignment as the fair treatment of claims. *Philosophical Studies*, 182(7): 1951–1973, 2025. doi: 10.1007/s11098-025-02300-4.

R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.

E. Goffman. *Frame analysis: An essay on the organization of experience*. Harvard university press, 1974.

D. Hadfield-Menell, A. D. Dragan, P. Fisac, and S. Russell. Cooperative inverse reinforcement learning. In D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, pages 3909–3917, 2016.

D. Hadfield-Menell, A. D. Dragan, and S. Russell. The off-switch game. In *Proceedings of the 31st AAAI Conference on Artificial Intelligence (AAAI 2017)*, pages 220–227, 2017.

J. Haidt. *The righteous mind: Why good people are divided by politics and religion*. Vintage, 2012.

C. Hammond, H. Bergman, and P. Brown. Pathological synchronization in parkinson's disease: networks, models and treatments. *Trends in neurosciences*, 30(7):357–364, 2007.

L. Hammond, A. Chan, J. Clifton, J. Hoelscher-Obermaier, A. Khan, E. McLean, C. Smith, W. Barfuss, J. Foerster, T. Gavenčiak, et al. Multi-agent risks from advanced ai. *arXiv preprint arXiv:2502.14143*, 2025.

W. A. Hershberger. An approach through the looking-glass. *Animal Learning & Behavior*, 14(4):443–451, 1986.

I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner. $\beta$-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. In *International Conference on Learning Representations*, 2017. URL `https://openreview.net/forum?id=Sy2fzU9gl`.

J. H. Holland. *Adaptation in Natural and Artificial Systems: An Introductory Analysis with Applications to Biology, Control, and Artificial Intelligence*. University of Michigan Press, Ann Arbor, MI, 1975. ISBN 0472084607.

L. Hong and S. E. Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.

E. Hutchins. *Cognition in the Wild*. MIT Press, Cambridge, MA, 1995.

M. Jaderberg, W. M. Czarnecki, I. Dunning, L. Marris, G. Lever, A. G. Castaneda, C. Beattie, N. C. Rabinowitz, A. S. Morcos, A. Ruderman, et al. Human-level performance in 3d multiplayer games with population-based reinforcement learning. *Science*, 364(6443):859–865, 2019.

I. L. Janis. *Victims of groupthink: A psychological study of foreign-policy decisions and fiascoes.* Houghton Mifflin, 1972.

H. J. Jeon, S. Milli, and A. Dragan. Reward-rational (implicit) choice: A unifying formalism for reward learning. *Advances in Neural Information Processing Systems*, 33:4415–4426, 2020.

Y. J. John, L. Caldwell, D. E. McCoy, and O. Braganza. Dead rats, dopamine, performance metrics, and peacock tails: Proxy failure is an inherent risk in goal-oriented systems. *Behavioral and Brain Sciences*, pages 1–68, 2023.

M. K. Johnson, S. Hashtroudi, and D. S. Lindsay. Source monitoring. *Psychological bulletin*, 114(1):3–28, 1993.

S. Johnson. *Where Good Ideas Come From: The Natural History of Innovation*. Riverhead Books, New York, 2010.

T. B. Kashdan and J. Rottenberg. Psychological flexibility as a fundamental aspect of health. *Clinical psychology review*, 30(7): 865–878, 2010.

S. A. Kauffman. The origins of order: Self-organization and selection in evolution. In *Spin glasses and biology*, pages 61–100. World Scientific, 1992.

R. Kegan. *The evolving self: Problem and process in human development*. Harvard University Press, 1982.

H. R. Kirk, I. Gabriel, C. Summerfield, B. Vidgen, and S. A. Hale. Why human–ai relationships need socioaffective alignment. *Humanities and Social Sciences Communications*, 12:728, 2025. doi: 10.1057/s41599-025-04532-5.

V. Krakovna, L. Orseau, R. Kumar, M. Martic, and S. Legg. Penalizing side effects using stepwise relative reachability. *arXiv preprint arXiv:1806.01186*, 2018.

V. Krakovna, A. Gleave, and J. Miller. Specification gaming: The flip side of AI ingenuity. DeepMind Safety Research Blog, May 2020. URL `https://www.deepmind.com/blog/specification-gaming-the-flip-side-of-ai-ingenuity`. Accessed on 2025-10-09.

T. S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, Chicago, 2nd edition, 1970. ISBN 9780226458083.

Z. Kurth-Nelson, T. Behrens, G. Wayne, K. Miller, L. Luettgau, R. Dolan, Y. Liu, and P. Schwartenbeck. Replay and compositional computation. *Neuron*, 111(4):454–469, 2023.

Z. Kurth-Nelson, S. Sullivan, J. Z. Leibo, and M. Guitart-Masip. Dynamic diversity is the answer to proxy failure. *Behavioral and Brain Sciences*, 47:e77, 2024.

B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30, 2017.

N. Lane and W. Martin. The energetics of genome complexity. *Nature*, 467(7318):929–934, 2010.

C. G. Langton. Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: nonlinear phenomena*, 42(1-3):12–37, 1990.

B. Latour. *We have never been modern*. Harvard university press, 2012.

D. Lazer and A. Friedman. The network structure of exploration and exploitation. *Administrative Science Quarterly*, 52(4):667–694, 2007.

J. Lehman and K. O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary computation*, 19(2):189–223, 2011.

J. Z. Leibo, A. S. Vezhnevets, W. A. Cunningham, S. Krier, M. Diaz, and S. Osindero. Societal and technological progress as sewing an ever-growing, ever-changing, patchy, and polychrome quilt. *arXiv preprint arXiv:2505.05197*, 2025.

J. Z. Leibo, A. S. Vezhnevets, M. Diaz, J. P. Agapiou, W. A. Cunningham, P. Sunehag, L. Cross, R. Koster, S. M. Bileschi, M. Chang, et al. A theory of appropriateness that accounts for norms of rationality. *arXiv preprint arXiv:2603.14050*, 2026.

N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

A. Livnat, C. Papadimitriou, J. Dushoff, and M. W. Feldman. A mixability theory for the role of sex in evolution. *Proceedings of the National Academy of Sciences*, 105(50):19803–19808, 2008.

A. Livnat, C. Papadimitriou, N. Pippenger, and M. W. Feldman. Sex, mixability, and modularity. *Proceedings of the National Academy of Sciences*, 107(4):1452–1457, 2010.

H. E. Longino. *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton University Press, Princeton, NJ, 1990.

J. Lorenz, H. Rauhut, F. Schweitzer, and D. Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.

L. Margulis and D. Sagan. *What Is Life?* Simon and Schuster, New York, 1995.

H. R. Maturana and F. J. Varela. *Autopoiesis and cognition: The realization of the living*. Springer Science & Business Media, 2012.

J. Maynard Smith. The origin and maintenance of sex. In *Group selection*, pages 163–175. Aldine Atherton, 1971.

D. J. McFarland and R. M. Sibly. The behavioural final common path. *Philosophical Transactions of the Royal Society of London B*, 270(907):265–293, 1975.

L. Messeri and M. J. Crockett. Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627(8002):49–58, 2024.

E. K. Miller and J. D. Cohen. An integrative theory of prefrontal cortex function. *Annual review of neuroscience*, 24(1):167–202, 2001.

G. A. Miller, G. Eugene, and K. H. Pribram. *Plans and the Structure of Behaviour*. Routledge, 1960.

S. Minuchin. *Families and Family Therapy*. Harvard University Press, 1974.

A. Moreno and M. Mossio. *Biological Autonomy: A Philosophical and Theoretical Enquiry*, volume 1 of *History, Philosophy and Theory of the Life Sciences*. Springer, Dordrecht, 2015. ISBN 978-94-017-9836-5. doi: 10.1007/978-94-017-9837-2.

H. Morrin, L. Nicholls, M. Levin, J. Yiend, U. Iyengar, F. DelGuidice, S. Bhattacharyya, J. MacCabe, S. Tognin, and R. Twumasi. Delusions by design? how everyday ais might be fuelling psychosis (and what can be done about it). *PsyArXiv*, 2025.

H. J. Muller. Some genetic aspects of sex. *The American Naturalist*, 66(703):118–138, 1932.

H. J. Muller. The relation of recombination to mutational advance. *Mutation Research*, 1(1):2–9, 1964.

T. Nagel. The fragmentation of value. In *Mortal Questions*, pages 128–141. Cambridge University Press, Cambridge, 1979.

T. Nagel. *The View from Nowhere*. Oxford University Press, New York, 1986. ISBN 9780195056440.

D. C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, Cambridge, 1990.

M. C. Nussbaum. *The fragility of goodness: Luck and ethics in Greek tragedy and philosophy*. Cambridge University Press, 2001.

S. Ohlsson. Information-processing explanations of insight and related phenomena. *Advances in the psychology of thinking*, 1: 1–44, 1992.

C. O'Keefe, P. Cihon, B. Garfinkel, C. Flynn, J. Leung, and A. Dafoe. The windfall clause: Distributing the benefits of ai for the common good. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, pages 327–331, 2020.

C. Olah, N. Cammarata, L. Schubert, G. Goh, M. Petrov, and S. Carter. Zoom in: An introduction to circuits. *Distill*, 5(3): e00024–001, 2020.

OpenAI. Pacing model development in an era of cyber-critical capabilities, 2026. URL `https://openai.com/index/pacing-model-development-cyber-capabilities/`. Accessed: 2026-08-21.

T. Ord. *The precipice: Existential risk and the future of humanity*. Hachette UK, 2020.

E. Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, 1990.

V. Padmakumar and H. He. Does writing with language models reduce content diversity? *arXiv preprint arXiv:2309.05196*, 2023.

S. E. Page. *The difference: How the power of diversity creates better groups, firms, schools, and societies*. Princeton University Press, 2008.

F. Perls, R. Hefferline, and P. Goodman. *Gestalt Therapy: Excitement and Growth in the Human Personality*. Julian Press, 1951.

M. Polanyi. *The Tacit Dimension*. Doubleday, Garden City, NY, 1966.

K. R. Popper. *Objective Knowledge: An Evolutionary Approach*. Clarendon Press, Oxford, 1972.

K. R. Popper. The rationality of scientific revolutions. In R. Harré, editor, *Problems of Scientific Revolution: Progress and Obstacles to Progress in the Sciences*, pages 72–101. Clarendon Press, Oxford, 1975.

I. Prigogine and I. Stengers. *Order Out of Chaos: Man's New Dialogue with Nature*. Bantam Books, New York, 1984.

J. Rawls. *Political Liberalism*. Columbia University Press, New York, 1993. ISBN 9780231052481.

H. R. Richardson and L. D. Stone. Operations analysis during the underwater search for scorpion. *Naval Research Logistics Quarterly*, 18(2):141–157, 1971.

W. J. Ripple and R. L. Beschta. Trophic cascades in yellowstone: the first 15 years after wolf reintroduction. *Biological Conservation*, 145(1):205–213, 2012.

N. Rosenberg. The direction of technological change: inducement mechanisms and focusing devices. *Economic Development and Cultural Change*, 18(1, Part 1):1–24, 1969.

S. Russell. *Human Compatible: Artificial Intelligence and the Problem of Control*. Penguin Publishing Group, 2019. ISBN 9780525558620. URL `https://books.google.co.uk/books?id=M1eFDwAAQBAJ`.

P. Scharre. *Army of None: Autonomous Weapons and the Future of War*. W. W. Norton, New York, 2018.

E. Schrödinger. *What is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, Cambridge, UK, 1944. Based on lectures delivered at Trinity College, Dublin, February 1943.

R. Shah, P. Freire, N. Alex, R. Freedman, D. Krasheninnikov, L. Chan, M. D. Dennis, P. Abbeel, A. Dragan, and S. Russell. Benefits of assistance over reward learning. *NeurIPS*, 2020.

M. Sharma, M. Tong, T. Korbak, D. Rogers, N. Bennett, Gabriel andjz, and E. Perez. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023.

N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

J. Sidanius and F. Pratto. *Social dominance: An intergroup theory of social hierarchy and oppression*. Cambridge University Press, 2001.

H. A. Simon. The architecture of complexity. *Proceedings of the American Philosophical Society*, 106(6):467–482, 1962.

S. Sontag, C. Drew, and A. L. Drew. *Blind man's bluff: The untold story of American submarine espionage*. Public Affairs, 1998.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.

Z. Sourati, A. S. Ziabari, and M. Dehghani. The homogenizing effect of large language models on human expression and thought. *Trends in Cognitive Sciences*, 2026.

D. Speijer, J. Lukeš, and M. Eliáš. Sex is a ubiquitous, ancient, and inherent attribute of eukaryotic life. *Proceedings of the National Academy of Sciences*, 112(29):8827–8834, 2015.

N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research*, 15(1):1929–1958, 2014.

K. O. Stanley, J. Lehman, and L. B. Soros. Open-endedness: The last grand challenge you've never heard of. O'Reilly Radar, 2017. URL `https://www.oreilly.com/radar/open-endedness-the-last-grand-challenge-youve-never-heard-of/`. Accessed: 2025-01-28.

P. D. Stokes. *Creativity from Constraint: The Psychology of Breakthrough*. Springer, New York, 2005.

C. R. Sunstein and R. Hastie. *Wiser: Getting Beyond Groupthink to Make Groups Smarter*. Harvard Business Review Press, Boston, MA, 2015.

J. Surowiecki. *The wisdom of crowds*. Vintage, 2005.

D. Susskind. *A World Without Work: Technology, Automation, and How We Should Respond*. Metropolitan Books, New York, 2020.

G. P. Wagner and L. Altenberg. Perspective: complex adaptations and the evolution of evolvability. *Evolution*, 50(3):967–976, 1996.

R. A. Watson and E. Szathmáry. How can evolution learn? *Trends in ecology & evolution*, 31(2):147–157, 2016.

A. Watts. *The Wisdom of Insecurity: A Message for an Age of Anxiety*. Vintage Books, New York, NY, 2011. Original work published 1951; Reference for The Backwards Law.

D. M. Wegner. Ironic processes of mental control. *Psychological review*, 101(1):34, 1994.

V. Weilnhammer, K. Y. Hou, R. Dolan, and M. M. Nour. Vulnerability-amplifying interaction loops: a systematic failure mode in ai chatbot mental-health interactions. *arXiv preprint arXiv:2602.01347*, 2026.

G. C. Williams. *Sex and Evolution*. Princeton University Press, Princeton, NJ, 1975.

L. Wittgenstein. *Philosophical Investigations*. Blackwell, Oxford, 1953.

A. W. Woolley, C. F. Chabris, A. Pentland, N. Hashmi, and T. W. Malone. Evidence for a collective intelligence factor in the performance of human groups. *Science*, 330(6004):686–688, 2010.

E. Yudkowsky. Coherent extrapolated volition. *Singularity Institute for Artificial Intelligence*, 2004.

K. J. S. Zollman. The epistemic benefit of transient diversity. *Erkenntnis*, 72(1):17–35, 2010.